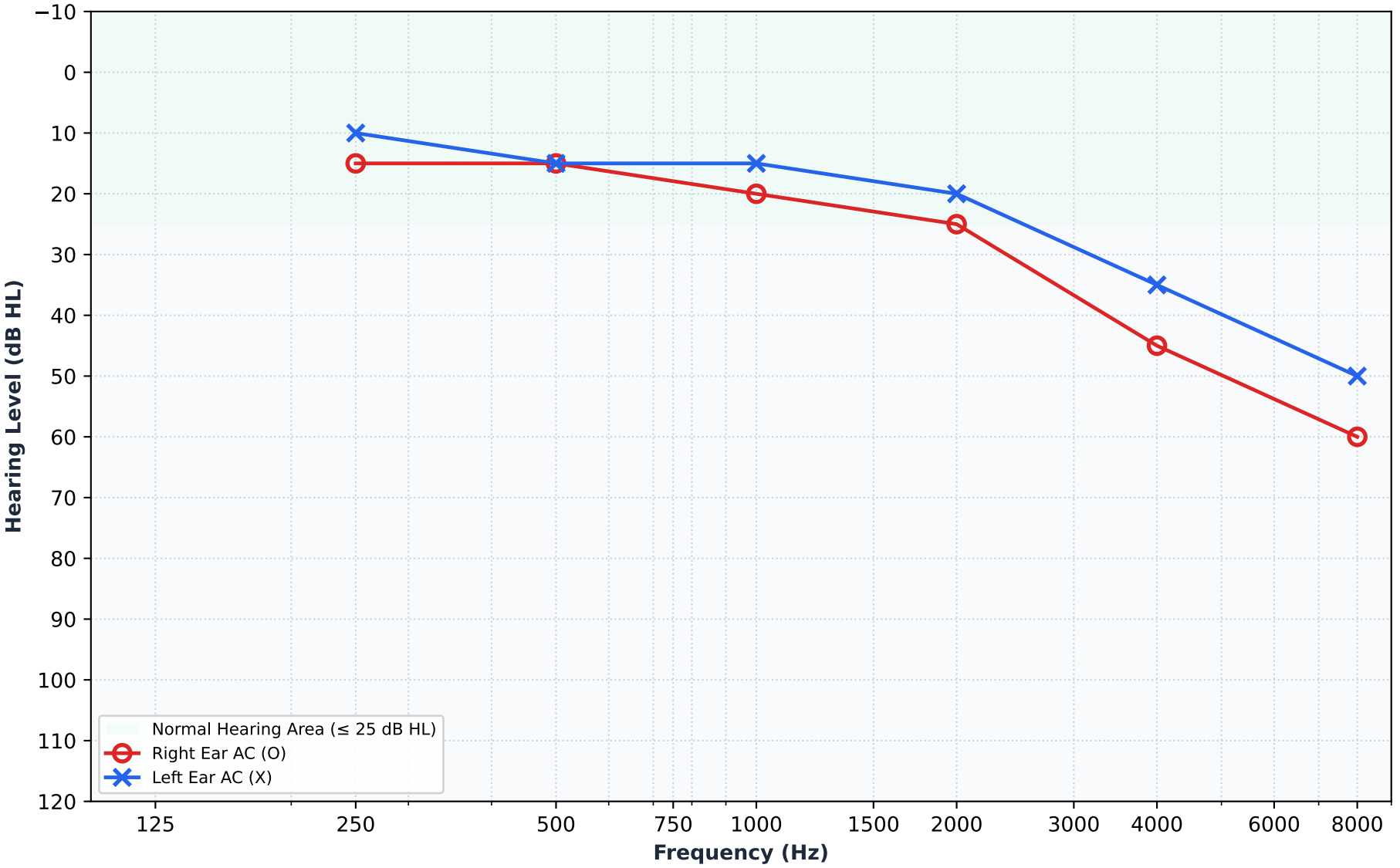
AUDIOLOGICAL EVALUATION & CLINICAL PATHOLOGY REPORT
Hardware-Grade Diagnostic Software Audiometer | ISO 8253-1 Compliant

Patient ID: PT-BENCH-99
Name: Test Subject
Age: 52 years
Date / Time: 2026-10-14T10:00

Clinician / Operator: Clinical Audiologist
Transducer Profile: Sennheiser_HDA200
Calibration Status: Calibrated (ISO 389-1)
Deployment: Rural Tele-Audiology Field Unit

PURE TONE AUDIOGRAM (BINAURAL)



Freq (Hz)	125	250	500	750	1000	1500	2000	3000	4000	6000	8000	PTA (4F)
Right AC (dB)	-	15	15	-	20	-	25	-	45	-	60	26.2 dB
Left AC (dB)	-	10	15	-	15	-	20	-	35	-	50	21.2 dB
Right BC (dB)	-	-	-	-	-	-	-	-	-	-	-	-
Left BC (dB)	-	-	-	-	-	-	-	-	-	-	-	-

SPEECH AUDIOMETRY SUMMARY
- Right Ear: SRT = 20 dB HL | WRS = 92%
- Left Ear: SRT = 15 dB HL | WRS = 96%
- Test Materials: Spondee & Phonetically Balanced Word Lists

RELIABILITY & QUALITY ASSURANCE
- Patient Reliability Score: 95.0%
- Ambient Room Noise Level: 36.5 dB SPL (ISO 8253-1 Permissible)
- Transducer Output Check: Digital Safety Limit Active (0 dBFS Clamping)

⚠ AC-ONLY INFERENCE — Approximate. Confirm with BC testing if available.
Diagnostic Classification: SENSORINEURAL (Certainty: 89.0%)
Clinical Recommendation: Refer for hearing aid evaluation and counseling.
Model Architecture: Random Forest Ensemble (AC_Only_Inference_v1) | Offline Local Inference